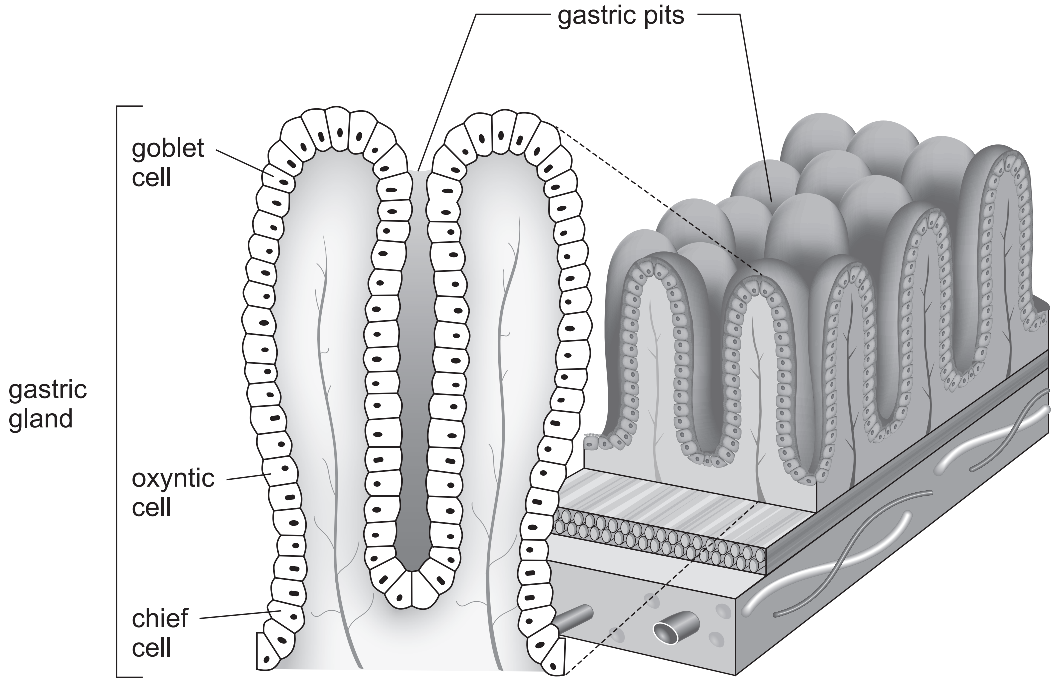
Examiner only

3. Gastric glands are responsible for producing a number of secretions. **Image 3.1** shows a section through the wall of the stomach.

Image 3.1



The chief cells produce and secrete pepsinogen, the inactive form of the endopeptidase pepsin. Oxyntic cells produce and secrete hydrochloric acid.

(a) (i) Explain why it is important that pepsin is produced in an inactive form **and** why it is necessary that hydrochloric acid and pepsinogen are produced by separate cells. [3]

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(ii) Describe the function of the goblet cells. [1]

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(iii) Glands within the wall of the duodenum produce exopeptidases.
Explain the advantage of the stomach producing endopeptidases and the duodenum producing exopeptidases. [2]

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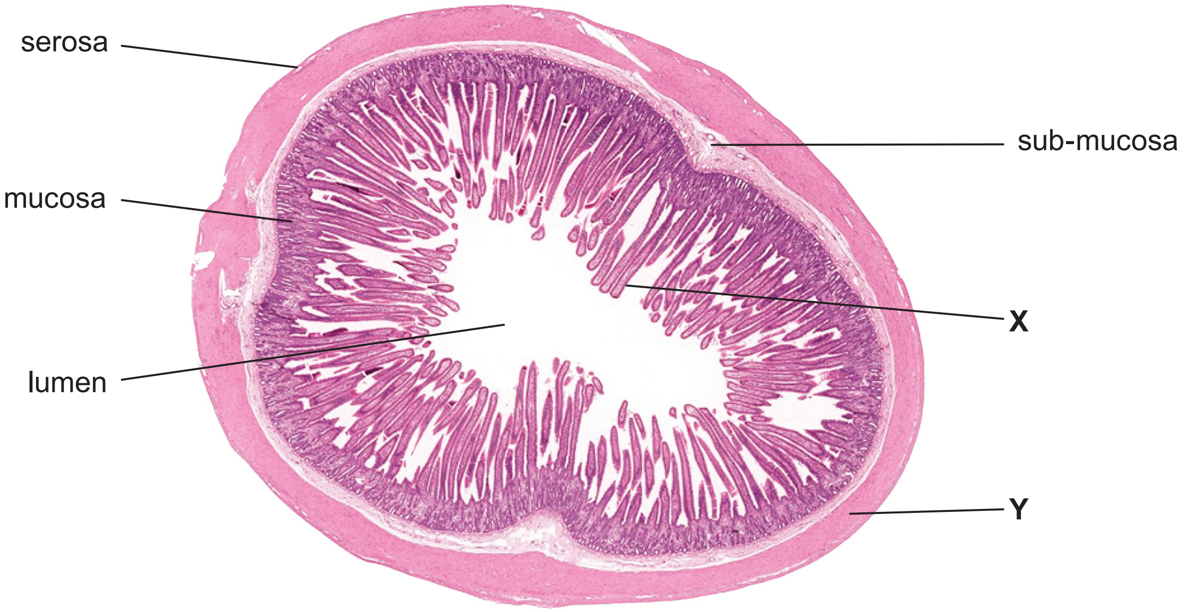
(b) **Image 3.2** shows a transverse section through the small intestine with some of the tissue layers labelled.

Image 3.2



Name the types of **tissue** found at [1]

X ....

Y ....

Examiner only

(c) Giardiasis is a common intestinal disease in some countries. It is caused by a parasitic protoctistan of the genus *Giardia*. The parasite causes damage to the cells in tissue **X** in **Image 3.2**, which result in shorter villi in the small intestine. The symptoms of infection include diarrhoea and fatigue (tiredness).

(i) State what is meant by the term parasite. [1]

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(ii) Explain how damage to the cells in tissue **X** can result in the symptoms described above. [4]

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